

PARIKSHIT

STUDENT SATELLITE

Seven subsystems. One student-built nanosatellite. Headed for orbit.

MANIPAL INSTITUTE OF TECHNOLOGY • MAHE • IN COLLABORATION WITH ISRO

MISSION OVERVIEW

Built by students. Bound for space.

Founded in 2013 at MIT Manipal, Parikshit is a student-built 2U-class nanosatellite developed for launch aboard an ISRO vehicle, with every subsystem designed, built and tested in-house.



Thermal Earth Imaging

Capture long-wave infrared images of the Indian subcontinent to monitor urban heat islands, ocean surface temperature and cloud cover.



Industry-Ready Students

Strengthen the link between students and the space industry through hands-on satellite development.



De-orbiting Innovation

Test an electrodynamic tether to actively de-orbit the satellite and reduce space debris after its mission life.

SEVEN SUBSYSTEMS, ONE SATELLITE

Every subsystem, engineered in-house.

Our 2U CubeSat is comprised of seven primary subsystems, each responsible for critical functions that ensure mission success and enable operation in the harsh environment of space.



01

ADCS

Orientation & control



02

COMMS

Communication link



03

EPS

Power generation



04

ODHS

Data handling



05

PAYLOAD

Mission instruments



06

STMS

Structure & Thermal



07

ADMIN & OPS

Outreach & Co-ordination

SUBSYSTEM 01

Attitude Determination & Control Subsystem

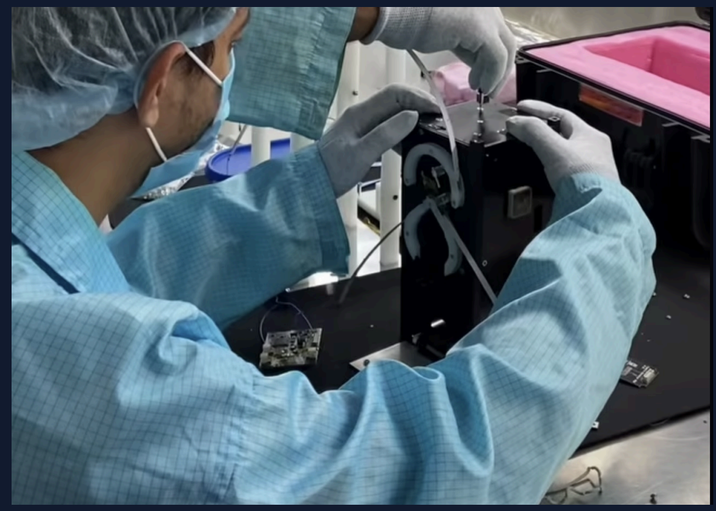
Responsible for determining and controlling the satellite's orientation in space.

KEY FUNCTIONS

1. Stabilizes the satellite after deployment from the launch vehicle
2. Utilizes sensors to determine the satellite's current orientation
3. Implements control algorithms to maintain desired orientation
4. Controls actuators for precise positioning

MAJOR COMPONENTS

Sun Sensors | Magnetometers | IMU | Magnetorquers | B-dot Controller | PD Controller | QuEST / EKF / MEKF | Orbit Propagator



SUBSYSTEM 02

Communications Subsystem

Serves as the primary space-to-ground link, enabling reliable telemetry, tracking and command (TT&C) operations.

KEY FUNCTIONS

1. RF Link Establishment — maintains high-reliability line-of-sight communication during ground station passes
2. Command Verification — receives, decodes and validates incoming uplink telecommands
3. Telemetry & Data Downlink — transmits real-time satellite health telemetry and payload data
4. Bandwidth & Link Optimization — maximizes spectral efficiency for the best link-budget performance

MAJOR COMPONENTS

- UHF/VHF Transceivers
- Amplifiers & Filters
- Antennas & Arrays
- Digital Modulation & Demodulation



COMMS

SUBSYSTEM 03

Electrical & Power Subsystem

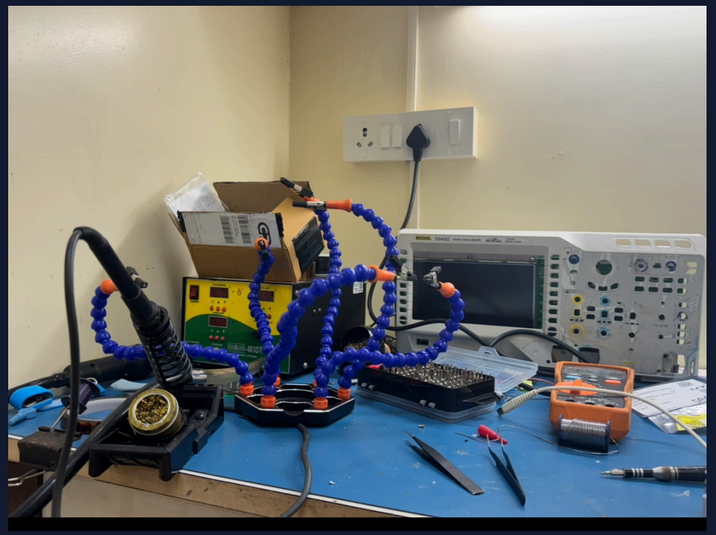
Generates, stores, and distributes electrical power to all satellite subsystems.

KEY FUNCTIONS

1. Generates power through high-efficiency solar panels
2. Stores energy in lithium-ion battery packs for eclipse periods
3. Regulates and distributes power to all subsystems
4. Implements power management algorithms to optimize usage
5. Provides protection against power anomalies and failures

MAJOR COMPONENTS

- Solar Panels (Triple-junction GaAs)
- Li-ion Battery Pack
- Load Switches
- DC-DC Converters
- Power Management Controller



EPS

SUBSYSTEM 04

Onboard Data Handling Subsystem

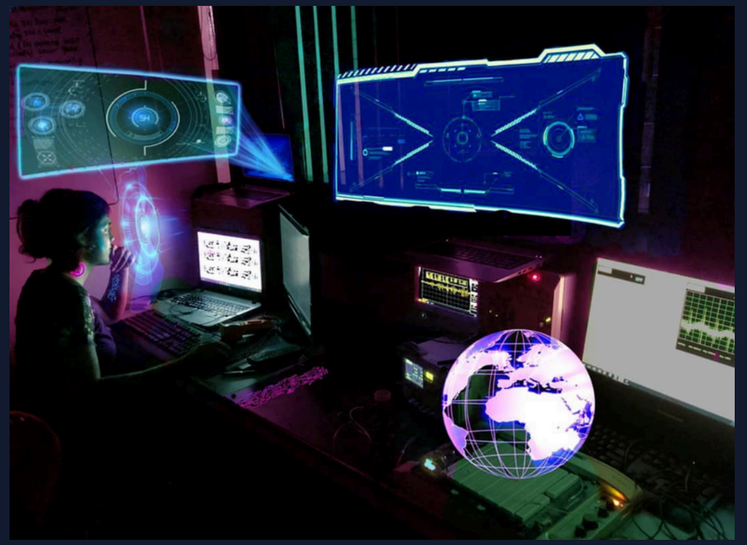
The brain of the satellite , processing commands and managing data flow between all subsystems.

KEY FUNCTIONS

1. Implements command processing and execution for satellite operations
2. Manages data collection, processing and storage
3. Handles system health monitoring and status reporting
4. Facilitates interboard communication and coordination of all software

MAJOR COMPONENTS

- OnBoard Computer
- Control State Machine
- Telemetry Storage
- Sensor Interfaces & Firmware
- Computer Architecture



ODHS

SUBSYSTEM 05

Payload Subsystem

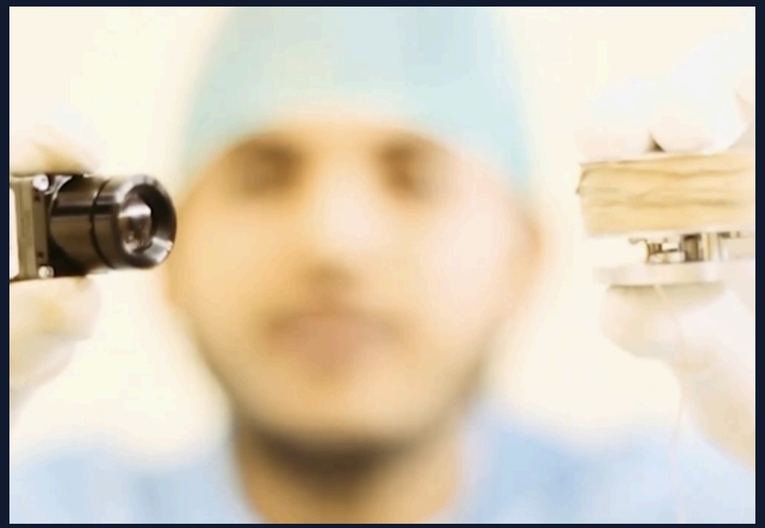
The functional part that delivers the satellite's purpose. The primary instruments it carries to perform its intended mission.

KEY FUNCTIONS

1. Manages primary mission instrumentation and sensors
2. Processes and stores scientific data collection
3. Controls experiment scheduling and execution
4. Monitors payload health and performance metrics
5. Maintains calibration of scientific instruments

MAJOR COMPONENTS

- Thermal Camera (Primary)
- Electrodynamic Tether (Secondary)



PAYLOAD

SUBSYSTEM 06

Structures, Thermal & Mechanisms Subsystem

Ensures the satellite's structural integrity, thermal stability, and reliable operation of its mechanical and deployment mechanisms throughout the mission.

KEY FUNCTIONS

1. Designs and manufactures the satellite structure to withstand launch forces
2. Maintains structural and thermal stability during space operations
3. Designs mechanical and deployment mechanisms for antennas and other deployables
4. Manages heat transfer via material selection, surface finishes and coatings

MAJOR COMPONENTS

- Aluminum Alloy Structure
- Deployment Mechanisms
- Mounting Rails & Brackets
- Solar Panel Mounts
- Passive Thermal Control
- Load-Bearing Interfaces



STMS

SUBSYSTEM 07

Admin & Operations

Keeps the team running and connected — managing the administrative functions of the satellite program.

KEY FUNCTIONS

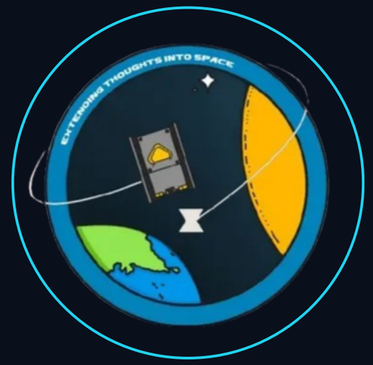
1. Website maintenance and updation
2. Reaches out to companies and sponsors for partnerships
3. Drives outreach for Parikshit inside and outside college
4. Handles social media pages and graphic design
5. Manages permissions and negotiations with affiliated organizations — university, space agencies, sponsors

FOCUS AREAS

Outreach & Representation

Operations & Coordination





Extending Thoughts Into Space

Seven subsystems. One student team. One nanosatellite headed for orbit.

PARIKSHIT STUDENT SATELLITE TEAM • MANIPAL INSTITUTE OF TECHNOLOGY, MAHE